

class-4

$i_{ac} \rightarrow$  generic AC wave form  
(square, sawtooth, sinusoidal)

$$P_{ac(av)} = \frac{1}{T} \int_0^T P(t) dt = \frac{1}{T} \int_0^T i_{ac}^2(t) \cdot R \cdot dt$$

$$\therefore P_{ac(av)} = P_{dc}$$

$$\Rightarrow \frac{1}{T} \int_0^T i_{ac}^2 \cdot R dt = I_{eff}^2 \cdot R$$

$$\Rightarrow I_{eff} = \sqrt{\frac{1}{T} \int_0^T i_{ac}^2(t) dt}$$

$$\Rightarrow I_{eff} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt} \rightarrow \text{Root Mean square (RMS)}$$

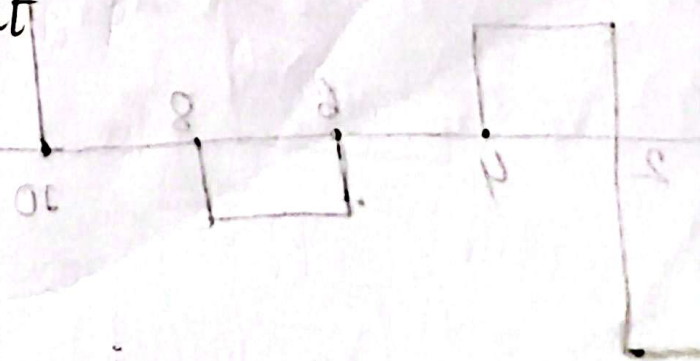
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$$x_{RMS} = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt}$$

$$V_{RMS} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

$$v(t) = V_m \sin \omega t$$

$$V_{RMS} = \frac{V_m}{\sqrt{2}}$$





## AC + DC waves

$$v_x(t) = v(t) + V_{DC} = V_m \sin \omega t + V_{DC}$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T v_x^2(t) dt} = \sqrt{\frac{1}{T} \int_0^T (v(t) + V_{DC})^2 dt}$$

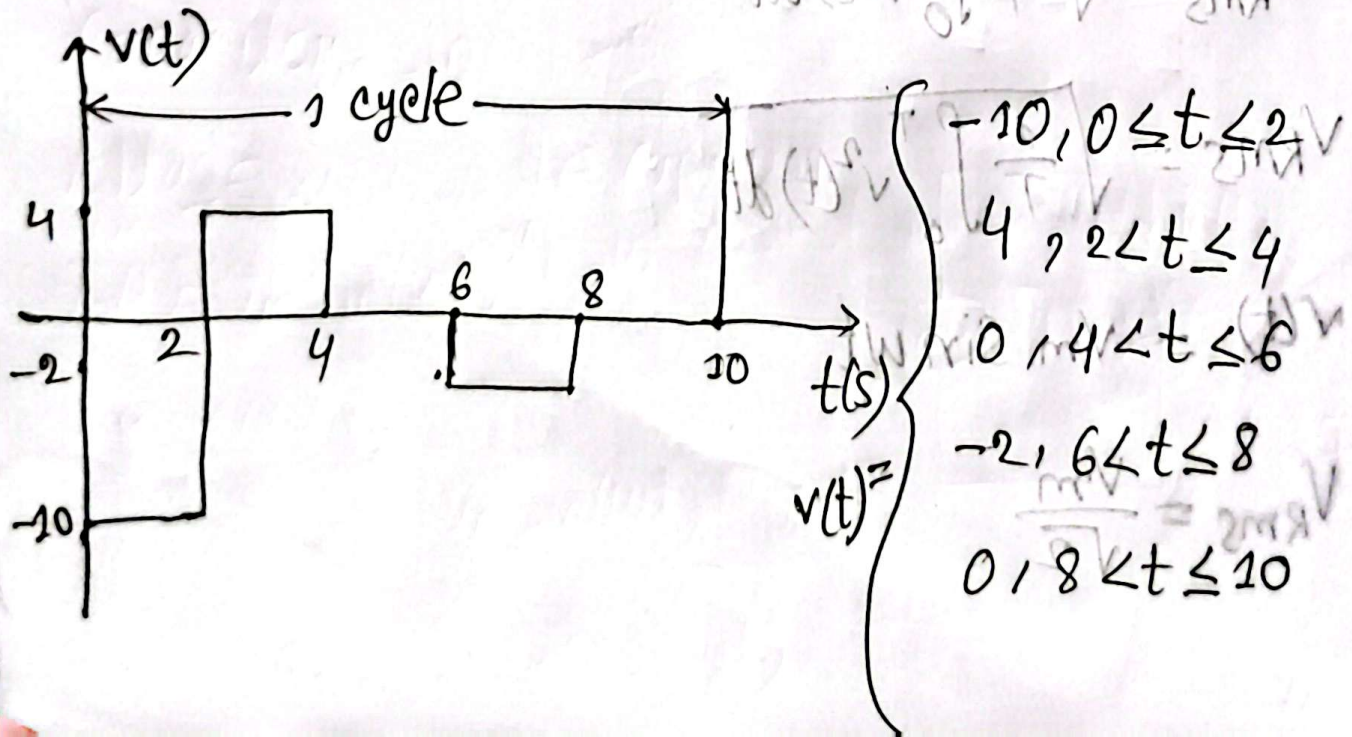
$$= \sqrt{\frac{1}{T} \int_0^T (v^2(t) + 2v(t)V_{DC} + V_{DC}^2) dt}$$

$$= \sqrt{\frac{1}{T} \int_0^T v^2(t) dt + \frac{2V_{DC}}{T} \int_0^T v(t) dt + \frac{V_{DC}^2}{T} \int_0^T dt}$$

$$= \sqrt{\left( \sqrt{\frac{1}{T} \int_0^T v^2(t) dt} \right)^2 + 0 + \frac{V_{DC}^2}{T} \cdot T}$$

$$\therefore V_{rms} = \sqrt{V_{ac}^2(RMS) + V_{DC}^2}$$

Q] Calculate the RMS value of the voltage.





$$V_{RMS} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

$$= \sqrt{\frac{1}{10} \times \left[ \int_0^2 (-10)^2 dt + \int_2^4 (4)^2 dt + \int_4^8 (-2)^2 dt \right]}$$

$$= \sqrt{\frac{1}{10} \times \left[ \int_0^2 100 dt + \int_2^4 16 dt + \int_4^8 4 dt \right]}$$

$$= \sqrt{\frac{1}{10} \left[ 100 \times (2-0) + 16 \times (4-2) + 4 \times (8-4) \right]}$$

$$= \sqrt{\frac{1}{10} \times 240} = 4.9V \text{ Ans}$$

# Response of Basic R, L, C elements to a sinusoidal voltage or current.

Resistor

voltage across the resistor,

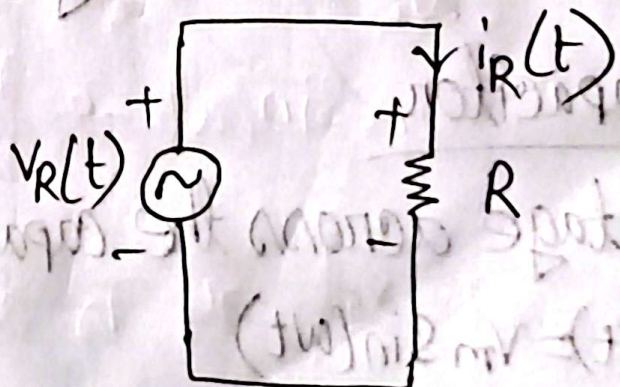
$$v_R(t) = V_m \sin(\omega t)$$

Applying ohm's law;

$$v_R(t) = i_R(t) * R$$

$$\Rightarrow i_R(t) = \frac{v_R(t)}{R}$$

$$\Rightarrow i_R(t) = \frac{V_m}{R} \sin(\omega t) \therefore i_R(t) = I_m \sin(\omega t)$$

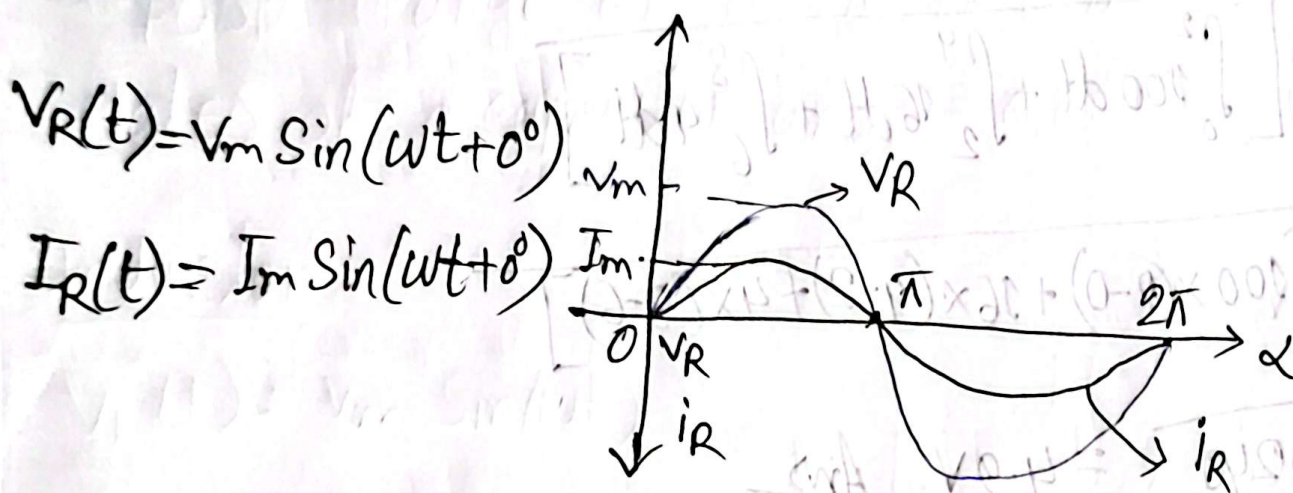




$$\therefore I_m = \frac{V_m}{R}$$

$$\Rightarrow \frac{V_m}{I_m} = R$$

Cause  
Effect = opposition



For a purely resistive element the voltage across and the current through the element are in phase, with their peak values related by ohm's law.

Capacitor

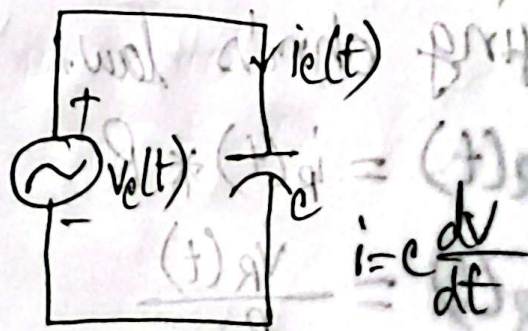
voltage across the capacitor.

$$V_C(t) = V_m \sin(\omega t)$$

we know,

$$i_C(t) = C \frac{d}{dt} [V_m \sin(\omega t)]$$

$$= C \cdot V_m \cdot \omega \cdot \cos(\omega t)$$





$$= \omega C V_m \sin(\omega t + 90^\circ) \quad [+ \cos \alpha = + \sin(\alpha + 90^\circ)]$$

$$= \frac{V_m}{\frac{1}{\omega C}} \sin(\omega t + 90^\circ)$$

$$= I_m \sin(\omega t + 90^\circ)$$

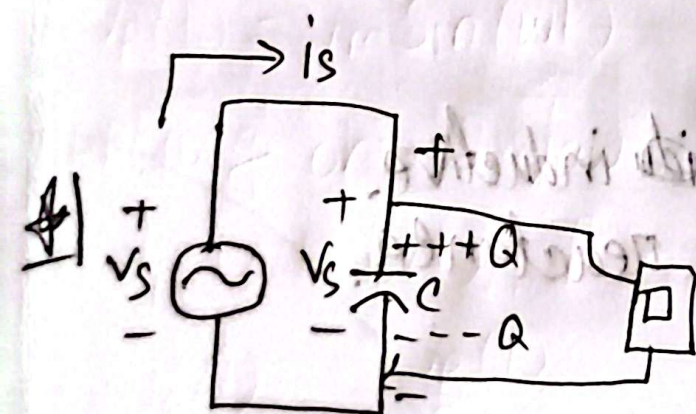
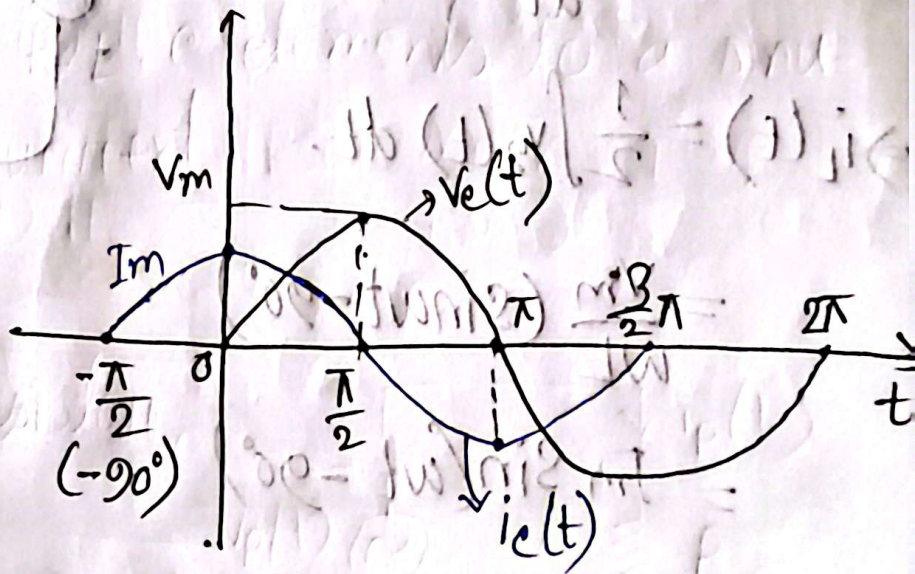
$$\therefore I_m = \frac{V_m}{\frac{1}{\omega C}}$$

$$\Rightarrow \boxed{\frac{V_m}{I_m} = \frac{1}{\omega C} = X_c} \rightarrow X_c \text{ is called capacitive reactance. Unit is ohm's.}$$

Cause = Opposition.  
Effect

$$V_c(t) = V_m \sin(\omega t + 0^\circ)$$

$$i_c(t) = I_m \sin(\omega t + 90^\circ)$$



$i_c$  leads  $V_c$  by  $90^\circ = \frac{\pi}{2}$

voltage  $\rightarrow$  charge  $\rightarrow$  Current.  
(from source)

~~need~~ this process for need voltage across capacitor.



An instantaneous change in voltage across a capacitor is opposed by the fact that there is an element of time required to deposit charge on (release ~~to~~ from) the plates of a capacitor and  $v = q/c$ .

## Inductor

$$V_L(t) = V_m \sin(\omega t)$$

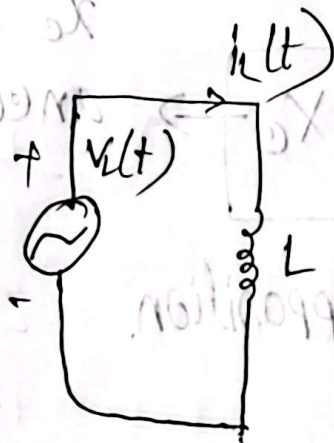
$$V_L(t) = L \frac{d}{dt}(i_L(t))$$

$$\Rightarrow i_L(t) = \frac{1}{L} \int V_L(t) dt$$

$$= \frac{V_m}{\omega L} (\sin \omega t - 90^\circ)$$

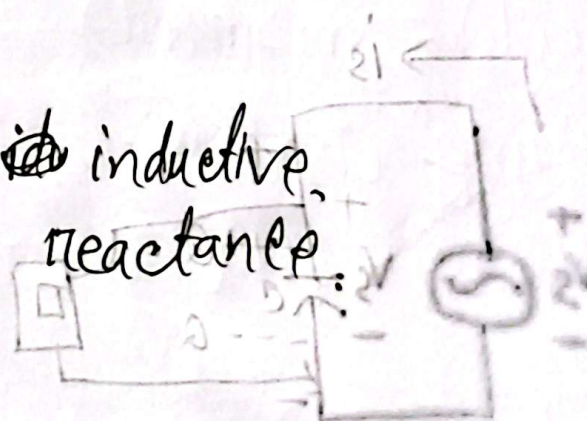
$$= I_m \sin(\omega t - 90^\circ)$$

$$\frac{V_m}{I_m} = \omega L = X_L, \quad X_L \text{ is called inductive reactance.}$$



$$V_L(t) = V_m \sin(\omega t - 90^\circ)$$

$$I_L(t) = I_m \sin(\omega t - 90^\circ)$$



Source (from) → voltage → change in current